XLVIII National Mathematics Olympiad: 3rd round, 17–18 April 1999

Problem 1. Find all triples (x, y, z) of natural numbers such that y is a prime number, y and 3 do not divide z, and $x^3 - y^3 = z^2$.

Problem 2. A convex quadrilateral of area S is inscribed in a circle whose centre is a point interior to the quadrilateral. Prove that the area of the quadrilateral whose vertices are the projections of the point of intersection of the diagonals on the sides does not exceed $\frac{S}{2}$.

Problem 3. In a competition 8 judges marked the contestants by *yes* or *no*. It is known that for any two contestants, two judges gave both a *yes*; two judges gave the first one a *yes* and the second one a *no*; two judges gave the first one a *no* and the second one a *yes*, and finally, two judges gave both a *no*. What is the greatest possible number of contestants?

Problem 4. Find all pairs (x, y) of integer numbers such that $x^3 = y^3 + 2y^2 + 1$.

Problem 5. Let B_1 and C_1 be points on the sides AC and AB of $\triangle ABC$. The straight lines BB_1 and CC_1 intersect at point D. Prove that the quadrilateral AB_1DC_1 is circumscribed if and only if the incircles of $\triangle ABD$ and $\triangle ACD$ are tangent.

Problem 6. Each interior point of an equilateral triangle of side 1 lies in one of six circles of the same radius r. Prove that $r \geq \frac{\sqrt{3}}{10}$.

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